

The Sharp Constant Two in Talponen's ODE-Determined $L^{p(\cdot)}$ Norms

Agent lane 15

June 29, 2026

Source and Target

The source paper is Jarno Talponen, *ODE to L^p norms*, arXiv:1402.0528. The paper defines an ODE-determined variable exponent norm by the terminal value $\|f\|_{p(\cdot)} = \varphi_f(1)$, where φ_f is the stabilized Caratheodory solution of

$$\varphi(0) = 0^+, \quad \varphi'(t) = \frac{|f(t)|^{p(t)}}{p(t)} \varphi(t)^{1-p(t)} \quad \text{for a.e. } t \in [0, 1].$$

After giving examples showing that the constant 2 would be best possible, the source asks whether

$$\frac{1}{2} \|f\|_1 \leq \|f\|_{p(\cdot)} \leq 2 \|f\|_\infty$$

always holds.

Proposition 3.1 (Hölder). *Suppose that $f \in L^{p(\cdot)}$ and $g \in L^{p^*(\cdot)}$ with $1 < p(t) < \infty$ for a.e. t . Then they satisfy Hölder's inequality:*

$$\int_0^1 |f(t)g(t)| \, dt \leq \|f\|_{p(\cdot)} \|g\|_{q^*(\cdot)}.$$

Although the function classes need not be linear spaces, we still use the norm notation above (instead of $\varphi_f(1)$ etc.); this is to establish a clear connection to the classical case.

Proof. By using the Hölder inequality for classical L^p and ℓ^p spaces, we obtain by induction an analogous statement for spaces of the type

$$(\dots (L^{p_1}(\mu_1) \oplus_{p_2} L^{p_2}(\mu_2)) \oplus_{p_3} \dots) \oplus_{p_n} L^{p_n}(\mu_n)$$

considered above. That is, if we write $\mu(A) = \sum_{i=1}^n \mu_i(A)$ and $f, g \in L^\infty(\mu)$ we have

$$\begin{aligned} \int |fg| \, d\mu &\leq \|f\|_{(\dots (L^{p_1}(\mu_1) \oplus_{p_2} L^{p_2}(\mu_2)) \oplus_{p_3} \dots) \oplus_{p_n} L^{p_n}(\mu_n)} \\ &\quad \cdot \|g\|_{(\dots (L^{p_1^*}(\mu_1) \oplus_{p_2^*} L^{p_2^*}(\mu_2)) \oplus_{p_3^*} \dots) \oplus_{p_n^*} L^{p_n^*}(\mu_n)}. \end{aligned}$$

This inequality passes to the limit by an approximation argument similar as seen previously. Namely, we pick by Lusin's theorem an increasing sequence of compact subset $C_n \subset [0, 1]$, $n \in \mathbb{N}$, such that $p|_{C_n}$ and $q|_{C_n}$ are continuous. Note that by the compactness of the subset and the continuity of the exponent we have $1 < \min_{t \in C_n} p(t) \leq \max_{t \in C_n} p(t) < \infty$ and similar holds for $q(\cdot)$. Then, similarly as in the proof of Theorem 2.3, for any sequence of simple semi-norms of standard form N_k with $\tilde{p}_{N_k} \rightarrow p(\cdot)$ in measure as $k \rightarrow \infty$ we have $N_k(1_{C_m} f) \rightarrow \|1_{C_m} f\|_{p(\cdot)}$ as $k \rightarrow \infty$. By essentially the same argument we also have that if N_k^* are the dual simple semi-norms of standard form, obtained by replacing all the exponents with their respective conjugates, then $\tilde{p}_{N_k^*} \rightarrow p^*(\cdot)$ in measure as $k \rightarrow \infty$ and we have $N_k^*(1_{C_m} g) \rightarrow \|1_{C_m} g\|_{p^*(\cdot)}$ as $k \rightarrow \infty$. This shows that

$$\int_{C_m} |fg| \, dt \leq \|1_{C_m} f\|_{p(\cdot)} \|1_{C_m} g\|_{p^*(\cdot)}.$$

Letting $m \rightarrow \infty$ yields the claimed inequality for L^∞ functions. Finally, by approximating the given functions $f \in L^{p(\cdot)}$ and $g \in L^{p^*(\cdot)}$ with functions of the form $1_D f, 1_D g \in L^\infty$ in measure and using the absolute continuity of the solutions φ_f and φ_g we obtain the statement. $\square$

Going back to simple semi-norms of standard form, note that if $p(\cdot) \equiv p_1$ on $[0, t_0]$ and $p(\cdot) \equiv p_2$ on $[t_0, 1]$ and $f \in L^{p(\cdot)}$ then $\|f\|_{L^{p(\cdot)}} = \|f\|_{L^{p_1}(\mu_1) \oplus_{p_2} L^{p_2}(\mu_2)}$ where $\text{supp}(\mu_1) = [0, t_0]$ and $\text{supp}(\mu_2) = [t_0, 1]$ (see Lemma 2.1).

It is easy to see that if $p_2 = 1$ then letting $p_1 \nearrow \infty$, $t_0 \searrow 0$ we have $\|1\|_{p(\cdot)} \nearrow 2$. This is perhaps surprising, since always $\|1\|_p = 1$ in the constant p case. We may also alter the above example as follows, letting above $f_{t_0} \equiv 1/t_0$ on $[0, t_0]$ and $f_{t_0} \equiv 1$ on $[t_0, 1]$ with $p_1 = 1$ and $p_2 \nearrow \infty$ and $t_0 \rightarrow 0^+$ yields that $\|f_{t_0}\|_{p(\cdot)} \rightarrow 1$ whereas $\|f_{t_0}\|_1 \rightarrow 2$.

We suspect that the above examples are characteristic in the sense that

$$\frac{1}{2} \|f\|_1 \leq \|f\|_{p(\cdot)} \leq 2 \|f\|_\infty$$

Idea of the Proof

The upper estimate improves the source paper's $e \|f\|_\infty$ bound. The source observes that after the profile reaches $\|f\|_\infty$, one has $\varphi' \leq \varphi$. But at and above that level the stronger estimate $\varphi' \leq \|f\|_\infty$ holds. Thus the profile can gain at most one more copy of $\|f\|_\infty$ during the remaining interval.

The lower estimate is most economical through Proposition 3.3 of the source paper, which compares the ODE norm to the corresponding Nakano norm. Young's inequality gives $\|f\|_1 \leq 2$ times the Nakano norm, and the Nakano norm is bounded above by the ODE norm.

Theorem 1. *Let $p : [0, 1] \rightarrow [1, \infty)$ be measurable and let $f \in L^{p(\cdot)}$ in Talponen's ODE-determined sense. Then*

$$\frac{1}{2} \|f\|_{L^1[0,1]} \leq \|f\|_{p(\cdot)} \leq 2 \|f\|_{L^\infty[0,1]},$$

with the usual interpretation that the upper inequality is vacuous if $\|f\|_\infty = +\infty$. The constant 2 is optimal on both sides.

Proof. The case $f = 0$ is trivial, so assume $f \neq 0$. Write $\lambda = \|f\|_{p(\cdot)} = \varphi_f(1)$.

First prove the upper bound. Put $M = \|f\|_\infty$. If $M = +\infty$ there is nothing to prove, and if $M = 0$ then $f = 0$, already excluded. Assume $0 < M < \infty$. If $\varphi_f(1) \leq M$, then $\varphi_f(1) \leq 2M$. Otherwise, by continuity and monotonicity of φ_f , choose $t_0 \in [0, 1]$ such that $\varphi_f(t_0) = M$. For almost every $t \in [t_0, 1]$ we have $\varphi_f(t) \geq M$ and $|f(t)| \leq M$, hence

$$\varphi'_f(t) = \frac{|f(t)|^{p(t)}}{p(t)} \varphi_f(t)^{1-p(t)} \leq \frac{M^{p(t)}}{p(t)} M^{1-p(t)} = \frac{M}{p(t)} \leq M.$$

Integrating over $[t_0, 1]$ gives

$$\|f\|_{p(\cdot)} = \varphi_f(1) \leq M + M(1 - t_0) \leq 2M.$$

Now prove the lower bound. Let

$$|||f|||_{p(\cdot)} = \inf \left\{ \mu > 0 : \int_0^1 \frac{1}{p(t)} \left(\frac{|f(t)|}{\mu} \right)^{p(t)} dt \leq 1 \right\}$$

be the Nakano norm used in Proposition 3.3 of the source paper. That proposition states that

$$|||f|||_{p(\cdot)} \leq \|f\|_{p(\cdot)}.$$

Let $\mu > |||f|||_{p(\cdot)}$ and set $u(t) = |f(t)|/\mu$. By the definition of the infimum, after decreasing μ slightly if necessary, we may assume

$$\int_0^1 \frac{u(t)^{p(t)}}{p(t)} dt \leq 1.$$

For every $q \geq 1$ and $s \geq 0$,

$$s \leq \frac{s^q}{q} + 1.$$

Indeed, this is immediate for $q = 1$, and for $q > 1$ it follows from Young's inequality $s \leq s^q/q + 1/q' \leq s^q/q + 1$. Applying this pointwise with $q = p(t)$ and $s = u(t)$, and using that the measure of $[0, 1]$ is one, we get

$$\frac{\|f\|_1}{\mu} = \int_0^1 u(t) dt \leq \int_0^1 \frac{u(t)^{p(t)}}{p(t)} dt + 1 \leq 2.$$

Letting $\mu \downarrow |||f|||_{p(\cdot)}$ yields $\|f\|_1 \leq 2|||f|||_{p(\cdot)} \leq 2\|f\|_{p(\cdot)}$, which is the desired lower bound.

The source paper’s two examples immediately before the question show sharpness. With p equal to p_1 on $[0, t_0)$ and 1 on $[t_0, 1]$, letting $p_1 \rightarrow \infty$ and $t_0 \rightarrow 0$ gives $\|\mathbf{1}\|_{p(\cdot)} \rightarrow 2 = 2\|\mathbf{1}\|_\infty$. The altered example $f_{t_0} = 1/t_0$ on $[0, t_0)$ and $f_{t_0} = 1$ on $[t_0, 1]$, with first exponent 1, second exponent tending to infinity, and $t_0 \rightarrow 0$, gives $\|f_{t_0}\|_{p(\cdot)} \rightarrow 1$ while $\|f_{t_0}\|_1 \rightarrow 2$. Hence no constant smaller than 2 can replace the right-hand constant, and no constant larger than 1/2 can replace the left-hand constant. $\square$

Verification and Novelty Notes

The proof is analytic. No numerical verification is used. The only source paper input beyond the definition of the ODE norm is Proposition 3.3, the comparison $|||f|||_{p(\cdot)} \leq \|f\|_{p(\cdot)}$.

Local duplicate checks were performed on June 29, 2026 in `registry_index.tsv`, `solutions/index.tsv`, `attempts/index.tsv`, and `proof_gaps/index.tsv` for arXiv id 1402.0528 and the phrases “ODE”, “variable exponent”, “constant 2”, “Nakano”, and variants of the displayed inequality. No existing packet for this target was found. A bounded web search for the arXiv id, the paper title, Talponen with “constant 2”, and exact text from the displayed inequality did not identify a later answer. This is a bounded-search novelty check, not a complete bibliographic audit.

Limitations and Human Review Recommendation

The proof answers the exact constant-two problem stated in the source paper. The reviewer should check only routine convention points: the interpretation when $\|f\|_\infty = +\infty$, the use of Proposition 3.3 for the source paper’s ODE-determined domain, and the endpoint $p(t) = 1$ in the elementary Young inequality step. I recommend human review as a likely valid full solution.

References

- [1] J. Talponen, *ODE to L^p norms*, arXiv:1402.0528.